THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

An Autonomous UAV for Post-Disaster Search, Localisation and Payload Delivery

First aircraft

1. Executive summary

Floods displace millions of people in India every year. The hard part of a rescue is not reaching survivors; it is finding them.

We want to build one small aircraft, about a metre across and about 7 kg, that searches flooded ground on its own. One person sends it out. It flies itself, spots people from the air, works out where those people are to within about a metre, and drops a relief kit to each. Nothing depends on a phone network or the internet.

If this aircraft shows the results we expect, we would build two more like it and fly all three together as a coordinated group. The design is already a three-aircraft design and the coordination software is already written and tested in simulation; what we are asking for here is the first aircraft, and the evidence that it works before anything is replicated.

The design and the software are done, and we have flown the whole mission in simulation. Nothing has flown for real yet, so every figure here is calculated rather than measured.

We would like to begin with INR 3,733: enough to test whether one motor lifts what the design needs. After that the money comes in 13 small steps, each released only once the step before it has produced a result we can show you. Nothing over INR 26,000 is spent until that first thrust measurement exists. If every step is approved the first aircraft comes to INR 2,45,993.

2. The problem

Boats are slow, dangerous for the crew, and cannot see onto rooftops. Helicopters are few and are needed for evacuation. Ordinary drones need a pilot each, rely on infrastructure the flood has washed away, and cannot help anyone once they have found them.

Finding people turns out to be the harder half. Someone in floodwater shows only their head and shoulders — about 40 cm of them is visible from above. Detection software works on a small picture of fixed size, so the usual approach shrinks the photograph, and the person shrinks with it until there is too little left to recognise. Using a smaller, cheaper model makes that worse, not better.

There is also no public set of labelled flood photographs to learn from. Thermal cameras do not rescue us either: someone sitting in water ends up close to the temperature of the water.

3. Our solution

The operator draws a search area of any shape and the software plans a route over it. We have tried this on thirteen awkward shapes and no route ever wandered outside the area. After that the aircraft flies itself, and our tests confirm nothing is commanded from the ground once a mission has begun. It spots people and works out where they are on its own, carries four relief kits, descends over a survivor, holds still, and drops one. If the radio fails, the battery runs low, or the operator hits abort, it ends the same way: come home and land.

The trick with the camera is not to shrink the picture. We cut each frame into tiles and search them at full size,

and we fly lower. Between them those two changes make a person in the water nine times bigger in the image, which is what lifts them clear of the range where this kind of software stops working. Neither change is enough on its own. It costs four times as much computing, which the accelerator we have chosen can supply, and the mission takes no longer, because flying lower saves more climbing than the extra passes cost.

There is one thing we have not solved. We plan to look at each patch of ground twice a second, which the accelerator manages comfortably. To be really confident about a sighting we would want three looks a second, and that is half as much work again — more than we can count on. There may be a way round it: most of what the camera sees is plainly open water, so a quick, cheap check could throw those patches away and leave the expensive search to the rest. The danger is that if the quick check throws away a patch with a person in it, nothing tells us. No error, no warning, just someone never found. We will measure how often that happens against public data before we rely on it.

What one aircraft can and cannot do

This is worth being exact about, because it is the whole argument for building the other two later.

The rate at which ground is searched is a property of the camera and the flying speed, not of how many aircraft there are: about 1.4 hectares a minute. One aircraft therefore sweeps **three and a half hectares** in the same three minutes that three aircraft take to sweep ten. Asked to cover the full ten hectares alone it needs about eight minutes of sweeping rather than three, and a complete mission then runs to roughly thirteen minutes against a hover endurance of about fifteen — which leaves none of the reserve our own sizing rule requires.

So the first aircraft demonstrates the whole mission, end to end, over the three-and-a-half-hectare share it would fly as one of three. Every part of the system is exercised: survey planning, autonomous flight, detection, localisation, descent, release, and return. What it cannot demonstrate is ten hectares inside the time limit with a fuel reserve, and it cannot deliver more than the four kits it carries. **That is precisely what the second and third aircraft buy, and it is why we would rather prove the first one first.**

4. Deliverables

1. One aircraft flying the mission end to end under one operator.
2. A ground station that shows what is happening and, by design, cannot fly the aircraft. It is built for three feeds and will show one.
3. The autonomy software, with an automatic test suite behind it. The multi-aircraft coordination is part of it and is already tested in simulation.
4. Measured detection accuracy and a verdict on the two-stage filter.
5. A publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
6. Source code and results, published in either outcome.
7. A measured answer to the question of whether to build the other two.

5. Benefits

- Three and a half hectares searched in about three minutes, and what comes back is a set of coordinates a rescue team can act on, not a video feed somebody has to sit and watch.
- The whole thing costs less than an hour and a half of helicopter charter at Indian rates, travels by road, and is run by two people.
- A thrust stand, a ground station, a tested codebase and students who have built all of it — and all of that stays in the department afterwards.
- Every part but one comes from an Indian supplier, so nothing sits at customs and anything broken in testing is replaced within days.

- The department finds out what a flying aircraft actually costs and actually does before being asked to fund three of them.

6. Resources needed

Funding is requested in 13 small steps rather than as one grant. Each step buys one named set of parts and is released only once the previous step has produced a stated result. If a step fails, the programme stops there and nothing beyond it has been spent.

The order follows engineering dependency: one motor is measured before the other three are bought, and the aircraft is assembled, weighed and flown before the ground segment is completed around it.

Phase schedule

The third column is the amount released. The fourth is what must already be true before it is released.

#	Objective	INR	Released only after
1	Establish thrust and mass measurement capability	3,733	Approval
2	Measure one motor, propeller and speed controller on the stand	14,771	Instruments commissioned
3	Aircraft 1 — autopilot and control link	12,626	**Thrust verified**
4	Aircraft 1 — onboard computer and AI accelerator	31,402	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning: RTK rover, compass and ground base	49,384	Compute stack operating
6	Aircraft 1 — camera and the three radios	16,301	Position fix acquired
7	Aircraft 1 — airframe fabrication	18,886	Avionics integrated
8	Aircraft 1 — battery pack and power distribution	24,194	Airframe fabricated
9	Aircraft 1 — release servos and wiring	4,701	Pack bench-discharged
10	Aircraft 1 — propulsion completed	25,661	Drive train installed
11	Charging and manual override	27,448	Aircraft assembled and weighed
12	Field consumables and assembly materials	7,934	Charging verified
13	Ground station: video feed, data link, base mount	8,953	Aircraft flying
	Total	**2,45,993**	

Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Nothing in this schedule replicates anything: every line buys a part that goes into the one aircraft, an instrument that measures it, or the ground station it reports to.

Component selection

The schedule gives the amount *released* per phase. The tables below give the parts each phase *buys*. A released amount is the parts cost plus tax where tax is still owed, plus 15 % contingency. Across the programme that is INR 2,12,381 of parts against INR 2,45,993 released.

Every line below is a current listing from a named Indian supplier, priced and linked. The schedule, the tables and the approval letters are all generated from that one list, so they cannot drift apart — and they are generated from the same list as the three-aircraft version of this brief, so the two documents cannot disagree about a price either.

One line is an allowance rather than a listing, and is marked as such: the consumables. GST is charged only on lines quoted before tax; Indian retail prices already include it. The full parts list, with every listing linked, is submitted alongside this brief as a separate Bill of Materials.

Terms used below. *RTK*: satellite positioning corrected against a second receiver on a surveyed point, giving centimetre rather than metre accuracy. *6S*: a battery of six cells in series, about 22 V. *kgf*: thrust written as the weight it would lift. *TOPS*: trillion operations per second, the accelerator's throughput. *ESC*: the electronic speed controller between battery and motor. *BMS*: the board that balances and protects the cells — there is none in this build, and the note after the tables says why.

Measurement instruments — phases 1–2 INR 3,246 of parts

Item	Model	INR	Why this part, and from whom
Load-cell amplifier	7Semi HX711 24-bit ADC	104	Turns the load cell's signal into a reading we can log. *Robu*
Load cell	REES52 20 kg load cell + HX711 breakout	854	The motors ship without a thrust curve, so we measure it ourselves. Ready-made stands cost around ten times this. *Amazon India*
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	373	Holds the motor clear of the bench, so its own draught does not flatter the reading. *Flipkart*
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1,140	A carbon airframe is bolted, not glued. *OnlyScrews*
Threadlocker	Loctite 243, 50 ml	664	Stops bolts working loose under motor vibration. Medium strength, so a joint can still be undone. *Amazon India*
Throttle source	Servo tester / ESC consistency tester, PPM	111	A motor on a stand needs something to command it. The safety-pilot transmitter is not bought until phase 11, long after this test. *Robokits*

Per aircraft, avionics and sensing — phases 3–6 INR 1,00,515 of parts

Item	Model	INR	Why this part, and from whom
Flight controller	Holybro Kakute H7 V1.5, no Bluetooth	9,269	The autopilot: flies the aircraft and runs its safety behaviours. The same flight software as the larger board it replaces, at under half the price, and it stacks directly on the speed controller. *Indian Robo Store*
FC vibration mount	Glass-fibre anti-vibration shock absorber	161	Isolates the autopilot from propeller vibration, which otherwise confuses its sensors. *Robu*
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1,549	The autopilot ships without one and records nothing without it. This is a different slot on a different board from the computer's card above. *Robocraze*
Companion computer	Raspberry Pi 5, 4 GB	11,049	The computer the autonomy runs on. The accelerator and the camera both plug into it, so neither works without it. The 4 GB model: detection runs on the accelerator, so more memory would buy nothing. *Yantronix*
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	11,749	Runs the person-detection model fast enough to search at flying speed. *Robu*
Compute cooling	Official Raspberry Pi 5 active cooler	519	The hottest moment is the setup on the ground, before the propellers are turning and cooling it. *Robu*
Storage	SanDisk High Endurance 128 GB microSD	3,989	Records every frame and detection for a whole mission. Ordinary cards wear out at that rate. *Flipkart*

Item	Model	INR	Why this part, and from whom
GNSS RTK rover	GenX F9P Air: ZED-F9P L1/L2 + helix + QMC5883 compass, 20 g	15,392	Centimetre positioning on every aircraft. Receiver, antenna and compass in one 20-gram unit. The compass matters: the new autopilot has none. *Robokits*
GNSS RTK base	SmartElex ZED-F9P-04B RTK board, U.FL	17,965	The fixed ground receiver the three aircraft correct against. It is ours, so no internet connection is needed at the site. *Robocraze*
GNSS antenna, base	LS-125-A L1/L2/L5 survey antenna	8,123	The base's antenna. Every aircraft's accuracy is only as good as this one fixed point, so it gets the better antenna. *Robokits*
GNSS feeder, base	U.FL to SMA female pigtail, 10 cm	49	Adapts the base receiver's miniature socket to the antenna cable. *Robocraze*
GNSS cable, base	RG58 50 ohm, 1 m, TNC-M to SMA-M	150	Carries the signal from the antenna on its tripod down to the base receiver. Converts the antenna's TNC plug to SMA. *IndiaMART*
GNSS cable extension, base	RG58, 2 m, SMA-F to SMA-M	210	Extends that cable to 3 m, so the antenna can stand clear of the station. *IndiaMART*
RTK base radio	EByte E22-900T22D-V2, SX1262, 22 dBm	989	Sends the base's corrections to the aircraft on the same radio type they already carry. *Robu*
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	2,827	Measures true height above whatever is below — water, ground or a rooftop. The kit is released at 6 m, and the barometer measures height above the launch point, which over floodwater is a different thing. *Indian Robo Store*
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	6,799	The search sensor, chosen so a person in water is large enough in the picture to be found. *Robu*
Video transmitter	LB-Link BL-M8812EU2, RTL8812EU, 5 GHz 2T2R	1,849	Sends the live picture down as digital video, one channel per aircraft. The computer draws its detections onto the picture first, so the judges see what the aircraft sees. *Hubtronics*
Video antennas	6 dBi 5 GHz dipole + 165 mm U.FL lead, 2 off	150	Two per aircraft, one for each of the radio's two channels. *Robocraze*
Command receiver	ExpressLRS RX24T, 2.4 GHz	1,439	Carries the safety pilot's control and the aircraft's reporting on a single link, which saves a second radio. *Robu*
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	989	How the three aircraft tell each other what they have found. The most robust link on the aircraft, which is where survivor positions belong. *Robu*
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	3,939	Powers the computer. Deliberately oversized: losing it in flight is the failure this line exists to prevent. *Amazon India*
BEC, video radio	2-6S 5 V 5 A switching BEC	959	A separate supply for the video radio, which draws two amps or more at full power. Rated 5 A so it has margin rather than running at its limit. *Robu*
Video radio cooling	Aluminium heatsink 50 x 25 x 10 mm + 3010 5 V fan	122	The video radio overheats without a heatsink and fan at full power. *Robu*
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	279	A separate supply for the servos, so a jammed servo cannot disturb the autopilot. *Robu*

Per aircraft, airframe and drive — phases 7–10 INR 70,986 of parts

Item	Model	INR	Why this part, and from whom
Motors	Tarot TL96020, 5008, 340 KV, 4 off	18,812	Twelve. The first is measured on the stand before the other eleven are bought. *Robokits*
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair, 6 off	9,846	Sixteen for twelve positions. The item most often broken in flight testing, and the whole of our spares policy. *Robokits*
Speed controllers	Hobbywing XRotor FPV G2, 4-in-1, 65 A, 3-6S	6,500	One board drives all four motors, with more than twice the current the motors are expected to draw. Confirmed on the thrust stand before the other two are bought. *RC Mumbai*
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm, 2 off	4,518	The four arms. Carbon, and comfortably strong — the joints, not the tubes, are what limit the airframe. *Robu*
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	2,895	The arms are tubes; this is the plate they bolt to. One sheet yields the top and bottom of a centre section at this size. *Indian Robo Store*
Motor mounts	Tarot 25 mm motor mount, TL9602, 4 off	4,476	Joins motor to arm without machining the tube. This joint is the part that has to be right. *Robu*
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	3,012	Bolted rather than bonded, so an arm damaged in a heavy landing is swapped in the field instead of scrapping the frame. *Amazon India*
Landing gear	F450/F550 landing skid	549	Gives clearance underneath for the kit magazine, and a stance wide enough for an untidy landing. *Robu*
Suspension springs	5 mm stainless double torsion spring, 10 off	20	Let the legs absorb a hard landing rather than pass it into the arms and the battery. *IndiaMART*
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	949	Filament for the magazine, mounts and covers. Holds its shape in the sun, which cheaper filament does not. *Robu*
Release servos	MG90S mini servo, 180 deg, 4 off	596	Open the four kit stations. Metal gears, because a plastic one can drop a kit without anyone knowing. *Robocraze*
Battery pack	GenX 22.8V 6S 16000mAh 25C/40C HV LiPo, AS150	12,197	About 365 Wh, the smallest listed size that still keeps the reserve for three aircraft landing in turn once its own weight is counted. Bought rather than built, so no spot welder and no cell-level assembly. It is heavier than the pack it replaces, which the thrust stand has to cover. *Robokits*
Pack fusing	ANL fuse holder + 150 A ANL fuses	3,000	Protects against a short circuit without cutting power at full throttle. *Njour*
Pack retention	300 mm LiPo strap, reusable, 2 off	124	Holds the battery down. It is a fifth of the aircraft's weight, so a strap that lets go is not a small problem. *Robu*
Main leads	10 AWG ultra-flexible silicone wire	149	The main power cable, sized for the highest current the motors ever draw rather than the average. *Robu*
Power connectors	Amass AS150 anti-spark, gold-plated, M/F pair, 4 off	2,133	Anti-spark, so connecting a battery this size does not burn the contacts. AS150 to match the battery's own plug, and rated 150 A against a peak near 136 A. *zBotic*
Signal connectors	JST ZH 6-pin, 200 mm leads	23	Keyed, so the wiring can only go back together one way after a repair. *Robu*
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1,139	Let the antennas mount on the airframe instead of hanging off the radios. *Desertcart*

Item	Model	INR	Why this part, and from whom
Insulation	PVC heat-shrink sleeve, 150 mm	47	Sleeving over the battery joints — the one place on the aircraft where a rubbed wire is an immediate fire. *Robu*

Flight-test support — phases 11–12 INR 29,867 of parts

Item	Model	INR	Why this part, and from whom
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	16,519	Manual override for the safety pilot, independent of the autonomy. A rule requirement and a safety one. *zBotic*
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	7,349	Two batteries recharged between flights is what sets how many tests fit into a day. *Robu*
Cable management	Nylon zip ties, 350 mm, 100 pcs	135	Ties long enough to dress wiring around a one-metre airframe. *Robu*
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	159	Insulates the signal wiring during assembly. *Robu*
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	560	Holds radios and receivers where a screw would need a fixing the printed part cannot carry. *Amazon India*
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	145	For the battery and anything else that must come out between flights without tools. *Amazon India*
Consumables	Fasteners, tape, connectors, filament	5,000	Solder, abrasives, spare hardware, filament. A round figure, because itemising it would be false precision. *in-house*

Ground segment and statutory — phases 13 INR 7,767 of parts

Item	Model	INR	Why this part, and from whom
Video receivers	LB-Link BL-M8812EU2, RTL8812EU, one per aircraft channel	1,849	Three, one per aircraft's channel, so the ground station can show all three at once, which the rules require. Digital, so no capture hardware. *Hubtronics*
Receive panels	5.8 GHz 14 dBi directional panel, RP-SMA, 62x55 mm	566	One directional antenna per aircraft, pointed at it, for range. *FlyRobo*
Panel pigtails	U.FL to RP-SMA female pigtail, RG178	99	Connects each receiver to its panel. *IndiaMART*
Receive antennas	6 dBi 5 GHz dipole + U.FL lead, second port	75	A second, all-round antenna on each receiver, which holds the link close in where the panel's narrow beam is easy to miss. *Robocraze*
Receiver USB hub	Transcend HUB3 powered 4-port USB hub + adapter	2,653	Powers the three ground receivers, which together draw more than a laptop's ports can supply. *Flipkart*
Coordination base	EByte E22-900T22U, SX1262, USB	1,646	The ground end of the link the aircraft use to report what they find. *Hubtronics*
Base station mount	Photographic tripod and adapter	879	Holds the ground receiver still during a mission. Survey-grade accuracy is unnecessary; not moving is what matters. *Amazon India*

Why there is no battery management board. An earlier version funded one. Most of what that line covered is already bought under other names, and the part that remains — a board that cuts power when a cell is unhappy — is the wrong device on an aircraft, because it can cut power at full throttle, which is a crash rather than a protection. Instead the battery is fused against short circuits, the autopilot brings the aircraft home on low voltage, and every cell is checked on the charger before each flight. What we do not have is per-cell protection in the air; that is a deliberate choice, not an oversight.

What this arrangement costs us

We should be honest that thirteen approvals is a lot of paperwork for the office, and that buying parts for one aircraft means single-unit prices with no quantity break. If the paperwork becomes the bigger problem, the per-subsystem steps can be merged into fewer orders.

The detection work needs no money at all: about a week on a departmental GPU and 30 GB of storage, against a dataset that is already public.

If the first aircraft works

Two more aircraft would come to about INR 3,32,186 — the remainder of the INR 5,78,179 three-aircraft programme — in sixteen further steps of the same kind. **We are not asking for that now, and we would not ask for it until the first aircraft has flown a complete mission in front of you.** Two things make the replication cheaper per aircraft than the first: the instruments, the charger, the safety-pilot transmitter and most of the ground station are bought once and are already in this ask, and the RTK base station bought with this aircraft serves all three.

7. Future work

Near term. Measure what is currently calculated: detection accuracy, motor thrust, setup time. Fly the full mission on one aircraft over its three-and-a-half-hectare share. Evaluate near-infrared sensing — water absorbs strongly and skin does not, so a person appears bright against dark water. A camera without its infrared filter costs INR 3,000, and a public dataset allows evaluation before purchase. Only the training data is flood-specific; the same system suits earthquake, wildfire and avalanche search.

Then, on evidence. Build the second and third aircraft and fly ten hectares in under eight minutes as a coordinated group. The coordination, launch deconfliction and area-splitting software this needs is already written and already tested in simulation with three aircraft; what it has never had is three real ones.

Longer term. A capable aircraft deciding for itself within a small power budget is a hard requirement in planetary aviation. *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density; *Dragonfly* launches in 2028 for Titan. A command takes minutes to arrive, so every decision is made on board. What transfers is what is not flood-specific: onboard survey planning, perception run to an energy budget, and safe behaviour without a link. Microcontroller-class processors, often proposed for such missions, cannot resolve targets of this size, so a planetary configuration needs a different operating point.

8. Conclusion

One aircraft, one operator, no network: people found in flooded ground and reached with a relief kit, and coordinates a rescue coordinator can act on. The design is finished. What is left is to build it and find out whether it works.

The department is never exposed to more than one step at a time. The first two come to INR 18,504 and settle whether these motors will lift this aircraft. If they will not, we stop there and the department still has a thrust stand. Nothing above INR 26,000 is released until that measurement exists.

And if the aircraft does what we expect it to, we would come back and ask to build two more, so that ten hectares can be searched in eight minutes with a reserve in hand — which is the one thing a single aircraft cannot do. That request would arrive with a flown mission behind it rather than a calculation.

Floods come back every monsoon, and nothing of this kind exists at a size two people can carry to the water's edge. We would like to begin, and we are asking the department to release Phase 1.

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-1

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 1 of 13, Establish thrust and mass measurement capability

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 1 of 13: Establish thrust and mass measurement capability. This is the first step, and is gated only on this approval. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1	1,140
Load cell	REES52 20 kg load cell + HX711 breakout	1	854
Threadlocker	Loctite 243, 50 ml	1	664
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	1	373
Throttle source	Servo tester / ESC consistency tester, PPM	1	111
Load-cell amplifier	7Semi HX711 24-bit ADC	1	104
		Total	**3,246**

We kindly request approval and financial support of **INR 3,246** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)	Signature	Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-2

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 2 of 13, Measure one motor, propeller and speed controller on the stand

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 2 of 13: Measure one motor, propeller and speed controller on the stand. It falls due only once the previous step has produced its stated result: *instruments commissioned*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Hobbywing XRotor FPV G2, 4-in-1, 65 A, 3-6S	1	6,500
Motors	Tarot TL96020, 5008, 340 KV	1	4,703
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	1	1,641
		Total	**12,844**

We kindly request approval and financial support of **INR 12,844** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)	Signature	Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-3

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 3 of 13, Aircraft 1 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 3 of 13: Aircraft 1 – autopilot and control link. It falls due only once the previous step has produced its stated result: *thrust verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Kakute H7 V1.5, no Bluetooth	1	9,269
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	1,549
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**10,979**

We kindly request approval and financial support of **INR 10,979** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)	Signature	Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-4

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 4 of 13, Aircraft 1 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 4 of 13: Aircraft 1 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Companion computer	Raspberry Pi 5, 4 GB	1	11,049
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**27,306**

We kindly request approval and financial support of **INR 27,306** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-5

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 5 of 13, Aircraft 1 – centimetre positioning: RTK rover, compass and ground base

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 5 of 13: Aircraft 1 – centimetre positioning: RTK rover, compass and ground base. It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS RTK base	SmartElex ZED-F9P-04B RTK board, U.FL	1	17,965
GNSS RTK rover	GenX F9P Air: ZED-F9P L1/L2 + helix + QMC5883 compass, 20 g	1	15,392
GNSS antenna, base	LS-125-A L1/L2/L5 survey antenna	1	8,123
RTK base radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
GNSS cable extension, base	RG58, 2 m, SMA-F to SMA-M	1	210
GNSS cable, base	RG58 50 ohm, 1 m, TNC-M to SMA-M	1	150
GNSS feeder, base	U.FL to SMA female pigtail, 10 cm	1	49
		Total	**42,878**

We kindly request approval and financial support of **INR 42,878** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)	Signature	Date

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-6

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 6 of 13, Aircraft 1 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 6 of 13: Aircraft 1 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	2,827
Video transmitter	LB-Link BL-M8812EU2, RTL8812EU, 5 GHz 2T2R	1	1,849
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
Video antennas	6 dBi 5 GHz dipole + 165 mm U.FL lead	2	150
Video radio cooling	Aluminium heatsink 50 x 25 x 10 mm + 3010 5 V fan	1	122
		Total	**14,175**

We kindly request approval and financial support of **INR 14,175** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)	Signature	Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-7

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 7 of 13, Aircraft 1 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 7 of 13: Aircraft 1 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	2,895
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	1	949
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	10	20
		Total	**16,419**

We kindly request approval and financial support of **INR 16,419** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
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Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-8

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 8 of 13, Aircraft 1 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 8 of 13: Aircraft 1 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Battery pack	GenX 22.8V 6S 16000mAh 25C/40C HV LiPo, AS150	1	12,197
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
BEC, video radio	2-6S 5 V 5 A switching BEC	1	959
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Pack retention	300 mm LiPo strap, reusable	2	124
		Total	**20,498**

We kindly request approval and financial support of **INR 20,498** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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To
The Dean, Student Affairs (DoSA)
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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-9

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 9 of 13, Aircraft 1 – release servos and wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 9 of 13: Aircraft 1 – release servos and wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Power connectors	Amass AS150 anti-spark, gold-plated, M/F pair	4	2,133
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Release servos	MG90S mini servo, 180 deg	4	596
Main leads	10 AWG ultra-flexible silicone wire	1	149
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**4,087**

We kindly request approval and financial support of **INR 4,087** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
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Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma	**Dr. Mamata Gulati**	**President, TAAS**
Mentor	Mentor	Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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The Dean, Student Affairs (DoSA)
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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-10

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 10 of 13, Aircraft 1 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 10 of 13: Aircraft 1 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	3	14,109
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	5	8,205
		Total	**22,314**

We kindly request approval and financial support of **INR 22,314** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR) _____ Signature _____ Date _____

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-11

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 11 of 13, Charging and manual override

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 11 of 13: Charging and manual override. It falls due only once the previous step has produced its stated result: *aircraft assembled and weighed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	1	16,519
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	1	7,349
		Total	**23,868**

We kindly request approval and financial support of **INR 23,868** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-12

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 12 of 13, Field consumables and assembly materials

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 12 of 13: Field consumables and assembly materials. It falls due only once the previous step has produced its stated result: *charging verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Consumables	Fasteners, tape, connectors, filament	1	5,000
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	1	560
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	1	159
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	1	145
Cable management	Nylon zip ties, 350 mm, 100 pcs	1	135
		Total	**5,999**

We kindly request approval and financial support of **INR 5,999** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)	Signature	Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-13

Date: _____

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 13 of 13, Ground station: video feed, data link, base mount

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 13 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 13 of 13: Ground station: video feed, data link, base mount. It falls due only once the previous step has produced its stated result: *aircraft flying*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Receiver USB hub	Transcend HUB3 powered 4-port USB hub + adapter	1	2,653
Video receivers	LB-Link BL-M8812EU2, RTL8812EU, one per aircraft channel	1	1,849
Coordination base	EByte E22-900T22U, SX1262, USB	1	1,646
Base station mount	Photographic tripod and adapter	1	879
Receive panels	5.8 GHz 14 dBi directional panel, RP-SMA, 62x55 mm	1	566
Panel pigtails	U.FL to RP-SMA female pigtail, RG178	1	99
Receive antennas	6 dBi 5 GHz dipole + U.FL lead, second port	1	75
		Total	**7,767**

We kindly request approval and financial support of **INR 7,767** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
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Approvals & Signatures

Dr. Surbhi Sharma Mentor	**Dr. Mamata Gulati** Mentor	**President, TAAS** Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

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Amount sanctioned (INR)	Signature	Date